TSIA2 Math Practice Test 14

PART A — QUESTIONS

TSIA2-M14-001
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A student compares two internet plans for online classes.

Plan A: $38 per month with no installation fee
Plan B: $24 per month plus a one-time $84 installation fee

For a 9-month school year, how much less does the cheaper plan cost?
A) $18
B) $24
C) $42
D) $126

TSIA2-M14-002
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
The formula $P = 2l + 2w$ gives the perimeter of a rectangle. Which equation correctly solves the formula for w?
A) $w = P - 2l$
B) $w = P/2 - l$
C) $w = (P - l)/2$
D) $w = 2P - l$

TSIA2-M14-003
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:
A patio is made from a 12-foot by 9-foot rectangle attached to a 6-foot by 6-foot square along one side. The two shapes do not overlap. What is the total area of the patio?
A) 90 square feet
B) 108 square feet
C) 132 square feet
D) 144 square feet

TSIA2-M14-004
Section: Math
Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
The table shows the results of a survey of 100 students.

Takes evening classes and works part time: 28
Takes evening classes and does not work part time: 12
Does not take evening classes and works part time: 36
Does not take evening classes and does not work part time: 24

If one student who works part time is chosen at random, what is the probability that the student takes evening classes?
A) 7/25
B) 7/16
C) 16/25
D) 7/10

TSIA2-M14-005
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A recipe for concrete mix uses 2.5 pounds of cement for every 4 pounds of sand. If a worker uses 30 pounds of sand, how many pounds of cement are needed?
A) 12.0 pounds
B) 18.75 pounds
C) 27.5 pounds
D) 48.0 pounds

TSIA2-M14-006
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
A taxi company uses the formula $C = 3.50 + 2.25m$, where C is the total cost in dollars and m is the number of miles. What does 2.25 represent in this formula?
A) The starting cost before any miles are traveled
B) The total cost of a 1-mile trip
C) The cost in dollars per mile traveled
D) The number of miles included in the starting cost

TSIA2-M14-007
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:
A scale model of a shipping crate has a volume of 40 cubic inches. The actual crate is similar to the model, and each actual length is 4 times the corresponding model length. What is the actual volume of the crate?
A) 160 cubic inches
B) 640 cubic inches
C) 1,600 cubic inches
D) 2,560 cubic inches

TSIA2-M14-008
Section: Math

Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
Two departments recorded the numbers of service requests completed by five employees.

Department A: 18, 20, 21, 22, 24
Department B: 12, 17, 21, 25, 30

Which statement best compares the two data sets?
A) Department A has a higher median and a smaller range.
B) Department B has a higher median and a smaller range.
C) Both departments have the same median, but Department A has a smaller range.
D) Both departments have the same median, but Department B has a smaller range.

TSIA2-M14-009
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A tank contains 18 liters of solution that is 40% cleaning concentrate. How many liters of cleaning concentrate are in the solution?
A) 4.5 liters
B) 7.2 liters
C) 10.8 liters
D) 22.5 liters

TSIA2-M14-010
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
A repair service charges according to the linear model shown in the table.

Hours: 1, 3, 5, 7
Total cost: \$85, \$145, \$205, \$265

Which equation represents the total cost C for h hours?
A) $C = 25h + 60$
B) $C = 30h + 55$
C) $C = 55h + 30$
D) $C = 60h + 25$

TSIA2-M14-011
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:
A line segment has endpoints (−2, 7) and (10, 1). What is the length of the segment?
A) $6\sqrt{3}$ units
B) $6\sqrt{5}$ units

C) $12\sqrt{2}$ units
D) 18 units

TSIA2-M14-012
Section: Math
Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
A student’s course grade is based on 60% exams and 40% assignments. The student has an exam average of 78. What assignment average is needed to have an overall course grade of 82?
A) 84
B) 86
C) 88
D) 90

TSIA2-M14-013
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A machine fills 360 bottles in 24 minutes. A second machine fills bottles at 80% of that rate. How many bottles does the second machine fill in 10 minutes?
A) 90 bottles
B) 120 bottles
C) 144 bottles
D) 180 bottles

TSIA2-M14-014
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
A company offers two payment options for a training course.

Option A: $320 up front
Option B: $95 registration fee plus $45 per week

After how many weeks will the two options cost the same?
A) 3 weeks
B) 4 weeks
C) 5 weeks
D) 6 weeks

TSIA2-M14-015
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:
Two parallel lines are cut by a transversal. One angle is labeled $(3x + 8)°$ and its corresponding angle is labeled $(5x − 22)°$. What is the measure of each of these corresponding angles?
A) 15°

B) 53°
C) 68°
D) 75°

TSIA2-M14-016
Section: Math
Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
A clinic tracks missed appointments over four months.

January: 18 missed appointments
February: 15 missed appointments
March: 12 missed appointments
April: 9 missed appointments

Which conclusion is supported by the data?
A) Missed appointments increased by 3 each month.
B) Missed appointments decreased by 3 each month.
C) The number of missed appointments stayed constant.
D) The total number of missed appointments was 45.

TSIA2-M14-017
Section: Math
Type: mcq
Category: Quantitative Reasoning
Prompt:
A rectangular room has a length of 15.8 feet and a width of 11.9 feet. Which is the best estimate of the room's area?
A) 160 square feet
B) 180 square feet
C) 200 square feet
D) 240 square feet

TSIA2-M14-018
Section: Math
Type: mcq
Category: Algebraic Reasoning
Prompt:
A ball is launched from the ground, and its height in feet after t seconds is modeled by $h(t) = -16t^2 + 48t$. For which positive value of t does the ball return to the ground?
A) 1 second
B) 2 seconds
C) 3 seconds
D) 4 seconds

TSIA2-M14-019
Section: Math
Type: mcq
Category: Geometric and Spatial Reasoning
Prompt:

Two similar rectangular signs have corresponding side lengths in a ratio of 3:7. The smaller sign has a perimeter of 48 inches. What is the perimeter of the larger sign?
A) 84 inches
B) 96 inches
C) 112 inches
D) 336 inches

TSIA2-M14-020
Section: Math
Type: mcq
Category: Probabilistic and Statistical Reasoning
Prompt:
A data set contains the values 8, 10, 11, 13, and 14. If the value 44 is added to the data set, which statement is true?
A) The mean increases more than the median increases.
B) The median increases more than the mean increases.
C) The mean decreases, and the median increases.
D) The mean stays the same, and the range decreases.

PART B — ANSWER KEY + EXPLANATIONS

TSIA2-M14-001 — Correct: C — Correct Answer: \$42 — Explanation: Plan A costs $9 \times \$38 = \$342$. Plan B costs $\$84 + 9 \times \$24 = \$300$. The cheaper plan costs $\$342 - \$300 = \$42$ less.
TSIA2-M14-002 — Correct: B — Correct Answer: $w = P/2 - l$ — Explanation: Starting with $P = 2l + 2w$, subtract $2l$ to get $P - 2l = 2w$. Divide by 2 to get $w = P/2 - l$.
TSIA2-M14-003 — Correct: D — Correct Answer: 144 square feet — Explanation: The rectangle has area $12 \times 9 = 108$ square feet. The square has area $6 \times 6 = 36$ square feet. The total area is $108 + 36 = 144$ square feet.
TSIA2-M14-004 — Correct: B — Correct Answer: 7/16 — Explanation: There are $28 + 36 = 64$ students who work part time. Of those, 28 take evening classes. The probability is $28/64 = 7/16$.
TSIA2-M14-005 — Correct: B — Correct Answer: 18.75 pounds — Explanation: The cement-to-sand ratio is 2.5/4. For 30 pounds of sand, the cement needed is $30 \times 2.5/4 = 18.75$ pounds.
TSIA2-M14-006 — Correct: C — Correct Answer: The cost in dollars per mile traveled — Explanation: In $C = 3.50 + 2.25m$, the coefficient of m is the rate of change, so 2.25 is the cost in dollars per mile.
TSIA2-M14-007 — Correct: D — Correct Answer: 2,560 cubic inches — Explanation: Volume scales by the cube of the length scale factor. Since each length is multiplied by 4, the volume is multiplied by $4^3 = 64$. The actual volume is $40 \times 64 = 2{,}560$ cubic inches.
TSIA2-M14-008 — Correct: C — Correct Answer: Both departments have the same median, but Department A has a smaller range. — Explanation: Both medians are 21. Department A's range is $24 - 18 = 6$, while Department B's range is $30 - 12 = 18$.
TSIA2-M14-009 — Correct: B — Correct Answer: 7.2 liters — Explanation: The amount of cleaning concentrate is 40% of 18 liters, or $0.40 \times 18 = 7.2$ liters.
TSIA2-M14-010 — Correct: B — Correct Answer: $C = 30h + 55$ — Explanation: The cost increases by \$60 when hours increase by 2, so the rate is \$30 per hour. Using $h = 1$ and $C = 85$ gives $85 = 30(1) + 55$, so $C = 30h + 55$.
TSIA2-M14-011 — Correct: B — Correct Answer: $6\sqrt{5}$ units — Explanation: The horizontal change is $10 - (-2) = 12$, and the vertical change is $1 - 7 = -6$. The distance is $\sqrt{(12^2 + (-6)^2)} = \sqrt{180} = 6\sqrt{5}$ units.
TSIA2-M14-012 — Correct: C — Correct Answer: 88 — Explanation: Let a be the assignment average. Then $0.60(78) + 0.40a = 82$. This gives $46.8 + 0.40a = 82$, so $0.40a = 35.2$ and $a = 88$.

TSIA2-M14-013 — Correct: B — Correct Answer: 120 bottles — Explanation: The first machine fills $360 \div 24 = 15$ bottles per minute. The second machine works at 80% of that rate, or 12 bottles per minute. In 10 minutes, it fills 120 bottles.
TSIA2-M14-014 — Correct: C — Correct Answer: 5 weeks — Explanation: Set the costs equal: $320 = 95 + 45w$. Then $225 = 45w$, so $w = 5$.
TSIA2-M14-015 — Correct: B — Correct Answer: 53° — Explanation: Corresponding angles are congruent, so $3x + 8 = 5x - 22$. Then $30 = 2x$, so $x = 15$. Each angle is $3(15) + 8 = 53°$.
TSIA2-M14-016 — Correct: B — Correct Answer: Missed appointments decreased by 3 each month. — Explanation: The values go from 18 to 15 to 12 to 9, decreasing by 3 each month.
TSIA2-M14-017 — Correct: B — Correct Answer: 180 square feet — Explanation: Estimate 15.8 as 16 and 11.9 as 12. Then $16 \times 12 = 192$, which is closest to 180 among the choices.
TSIA2-M14-018 — Correct: C — Correct Answer: 3 seconds — Explanation: Set $h(t) = 0$: $-16t^2 + 48t = 0$. Factor to get $-16t(t - 3) = 0$. The positive solution is $t = 3$.
TSIA2-M14-019 — Correct: C — Correct Answer: 112 inches — Explanation: Perimeter scales by the same factor as side length. The scale factor from smaller to larger is 7/3. The larger perimeter is $48 \times 7/3 = 112$ inches.
TSIA2-M14-020 — Correct: A — Correct Answer: The mean increases more than the median increases. — Explanation: The original mean is $56 \div 5 = 11.2$, and the new mean is $100 \div 6 \approx 16.7$, an increase of about 5.5. The original median is 11, and the new median is $(11 + 13)/2 = 12$, an increase of 1.